CTSA Human-Return Readout

A Session-Boundary Measurement Architecture for Retained Human Capability After AI

Programme continuation, Readout-native synthesis, and global-literature constraint

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Status. Final author draft of a K0 conceptual and measurement architecture. "Final" denotes the present manuscript state, not empirical validation, publication, or independent certification. The paper does not claim that Concept, Tool-Selection, Skill-Execution, and Alternative-Generation are four biological modules, exhaustive kinds of knowledge, or validated psychometric factors. It proposes four *readout classes* for describing what remains demonstrably usable in a human after AI support is removed. The external review is a targeted, publisher/DOI-verified integrative constraint review conducted for this revision; it is not a PRISMA systematic review. Internal programme manuscripts establish genealogy and conceptual continuity, not independent validation.

Central claim. Human-AI learning should not be inferred from what the coupled system produces while assistance is present. It should be tested at the session boundary: after AI removal, what readout-distinguishable capability is still available to the human, what previously available capability has been lost, and what epistemic status has the retained change earned? CTSA codes the *content* of retained change as Conceptual Return (C), Tool-Selection Return (T), Skill-Execution Return (S), or Alternative-Generation Return (A). It keeps persistence, loss, metacognitive regulation, provenance, warrant, and epistemic direction separately auditable.

1. Abstract

Human-AI collaboration is commonly evaluated at the moment of assistance: answer quality, speed, trust, preference, or joint performance. The Human-AI Readout Programme has progressively rejected that single-horizon view. *When AI Expands Human Potential* located the target in corrigible, world-responsive human agency rather than productivity; *Human Learning as Epistemic Architecture* connected concepts to problem-context, tools, workflow, skill, and real-world feedback; *From Problem to Hypothesis* modeled bounded, history-shaped semantic mobility and readout-discriminable alternatives; *Experience Is the Human LoRA* separated exposure from selectively retained adaptation; Readout Genesis placed retained difference before semantic naming and required explicit loss accounting; and *Epistemic Fusion or Tunnel* supplied the session architecture, candidate status, Difference-Resistance-Agency conditions, retention, epistemic direction, and an unaided Human Return battery [1, 2, 5, 3, 4, 6, 7].

The residual problem is measurement typing. The Human Return Test asks whether a person can reconstruct, discriminate, transfer, and formulate a better next question after AI is gone, but it does not compactly classify *what kind* of retained change is being demonstrated. This paper proposes **CTSA Human-Return Readout**: four observable readout classes—Conceptual Return, Tool-Selection Return, Skill-Execution Return, and Alternative-Generation Return—crossed with an unaided return test and accompanied by loss, metacognitive-regulation, provenance, warrant, and direction audits.

Global literature constrains the novelty claim. Conceptual, procedural, conditional, and metacognitive knowledge are established constructs; self-regulated learning already separates knowing what, how, and when/why; conceptual bootstrapping and curriculum-learning research show that structured components can accelerate later learning; Human-AI studies show that confidence calibration, uncertainty expression, verification cost, and cognitive forcing affect reliance; and recent GenAI education evidence directly separates assisted performance from later unaided learning [8, 9, 12, 13, 19, 20, 21, 22, 16, 17, 18]. Accordingly, CTSA does *not* propose four new kinds of knowledge. Its contribution is a session-boundary measurement architecture that asks whether a retained Human-AI change can be typed, demonstrated without decisive AI assistance, compared with a frozen pre-AI baseline, and interpreted without collapsing gain into net expansion or persistence into warrant.

2. The Problem: Assisted Output Is Not Yet Human Learning

The earliest programme question was not whether AI can make a person faster. It was whether AI-mediated inquiry can enlarge the human capacity to remain corrigible, compare alternatives, encounter resistant conditions, and revise judgment in ways that remain answerable to the world [1]. Fusion/Tunnel sharpened the target: repeated Human-AI dialogue should enlarge a human being's reachable, discriminable, corrigible space of inquiry rather than merely accelerate output or outsource epistemic work [6].

External evidence now makes the temporal distinction unavoidable. Bastani et al. found in a field experiment with nearly one thousand high-school mathematics students that GPT access improved performance during practice, yet the unguarded GPT condition performed worse than control after access was removed; learning-oriented safeguards largely mitigated the loss [16]. Yan et al. therefore argue that performance gains from GenAI must be distinguished from learning because high-quality learning depends on cognitive and metacognitive processing that fluent completion can bypass [17]. Deng et al.'s systematic review and meta-analysis of experimental ChatGPT studies found positive average effects across several educational outcomes, but also emphasized limitations in post-intervention assessment and the need for longer-horizon, skill-demonstrating measures [18].

The programme's existing non-collapse law is therefore not merely philosophical:

$$\text{Assisted performance} \neq \text{retained human capability}.$$

The practical measurement question becomes: *after assistance is removed, what, exactly, came back with the human?*

3. Programme Genealogy: Tight Internal Continuity

The proposed architecture is not a detached taxonomy added late to the programme. It is the missing interface among objects already developed in separate manuscripts.

The genealogy yields one narrow residual contribution. Human Learning already contains Concept, Tool, Workflow, and Skill; Problem-to-Hypothesis already contains discriminable alternatives; Human LoRA already contains selective persistence; Fusion/Tunnel already contains Human Return. What was missing was a compact, typed readout layer for the *content* of retained change at the Human-AI session boundary.

Residual contribution. CTSA does not claim that learning consists of four things. It proposes that, when a retained Human-AI change is assessed after AI removal, one useful readout profile is whether the human can now demonstrate a new Conceptual distinction, appropriately select a Tool/access route, execute and transfer a Skill, or independently generate a non-equivalent Alternative—with losses, regulation, provenance, warrant, and direction kept separately visible.

4. External Literature as Constraint, Not Validation

The external literature does two jobs. First, it prevents CTSA from renaming established constructs as discoveries. Second, it identifies the narrower interface that remains under-integrated.

Krathwohl's revision of Bloom's taxonomy distinguishes factual, conceptual, procedural, and metacognitive knowledge [8]. Wirth et al. review self-regulated learning as layered content, strategy, and metacognitive regulation, and explicitly distinguish declarative, procedural, and conditional strategy knowledge—what, how, and when/why [9]. These literatures block any claim that Concept, Tool, and Skill are unprecedented categories.

Zhao, Lucas, and Bramley show that human conceptual bootstrapping can reuse and recombine prior concepts to construct more complex ones [12]. Mi and Summerfield show that "divide and conquer" curricula can facilitate later performance on multi-cue combinations [13]. These findings support the plausibility of compositional and role-structured learning, but they do not establish that CTSA classes are the correct or exhaustive decomposition of Human Return.

Human-AI literature constrains the control layer. Ma et al. show that calibrating human self-confidence matters for appropriate reliance [19]; Kim et al. show that natural-language uncertainty expression can reduce agreement with an LLM and reduce overreliance on incorrect answers in some conditions [20]; Vasconcelos et al. show that verification behavior depends on task and explanation costs [21]; and Buçinca et al. show that cognitive forcing can reduce overreliance, though sometimes at a cost to preference [22]. CTSA must therefore distinguish *what returned* from whether the human can monitor, select, check, and revise that return.

Finally, Vaccaro, Almaatouq, and Malone's meta-analysis separates human augmentation from Human-AI synergy [23]; Doshi and Hauser show that generative AI can improve individual creativity while increasing similarity across outputs [24]; Vives et al. report a discrimination-generalization trade-off associated with more differentiated semantic representations [14]; and Holton et al. show transfer-interference trade-offs in continual learning [15]. Together these results motivate CTSA's strongest non-collapse: a gain in one observable dimension does not by itself establish net human epistemic expansion.

5. Readout-Native Foundations

5.1 Retained difference before semantic label

Readout Genesis places retention before human memory semantics and meaning after readout. A reader makes a distinction accessible under a question; it does not retroactively place that meaning in the root. Approximate bridges must expose named defects, and familiar semantic labels cannot be imported as root-level truths merely because they are intuitive [4].

CTSA therefore begins with a declared pre/post difference, not four assumed cognitive boxes. Let

$$S_H^{\text{pre}}, \qquad S_H^{\text{post}}$$

be bounded task-relative descriptions of the human before AI exposure and after an unaided return interval. A declared reader $\mathcal{R}_Q$ asks whether a task-relevant difference can be recovered:

$$\Delta_H^{\text{ret}} = \mathcal{R}_Q(S_H^{\text{post}}) - \mathcal{R}_Q(S_H^{\text{pre}}).$$

This is architectural notation, not a validated psychological metric. The reader does not discover a

Table 1: Internal programme genealogy. Each prior manuscript contributes a distinct object; self-citation establishes lineage, not independent validation.

Programme manuscript	Retained result	Function inside CTSA v9.1
When AI Expands Human Potential	AI can expose alternatives and revision pressure, but human potential requires corrigibility, route-openness, criticizability, and world-answerability.	Supplies the normative target: human epistemic development rather than convenience or output.
Human Learning as Epistemic Architecture	Learning runs from lived word/meaning through relation, retrieval, constraint and revision, then through problem-context, context-appropriate tools, workflow, practiced skill and feedback.	Supplies the concept-to-competence arc and the distinction between tool selection, workflow, and skill.
Experience Is the Human LoRA	Experience is not automatically learning; only selected residues persist, consolidate, transfer, or alter later interpretation/action.	Supplies the session-boundary retention requirement and the warning that exposure is not durable adaptation.
Readout Genesis / MEMK	Retained difference precedes semantic naming; meaning follows readout; approximate translation must expose loss; meaning, truth, and epistemic status remain distinct.	Prevents CTSA from becoming a root ontology and requires loss/claim-boundary accounting.
From Problem to Hypothesis	Semantic mobility is bounded and history-shaped; reachability differs from accessibility; idea count differs from readout-discriminable hypothesis classes.	Supplies Alternative Return as a genuinely non-equivalent accessible route and the no-silent-loss discipline.
Epistemic Fusion or Tunnel v7	Dialogue is a reciprocal, history-shaped semantic reaction; Difference, Resistance, and retained human Agency are non-substitutable for durable human epistemic expansion.	Supplies the Human-AI session architecture and the requirement that correction returns as human capacity.
Epistemic Fusion or Tunnel v8.1	Separates candidate status from validation, retention from direction, augmentation from synergy, and joint gain from delayed Human Return; operationalizes an unaided return battery.	Supplies the immediate measurement horizon into which CTSA is inserted: after retention, before the claim of human expansion.

complete human state; it checks whether a specified capability or distinction is demonstrably available under frozen criteria.

5.2 Meaning, persistence, and truth remain non-identical

Readout Genesis separates data from meaning, meaning from truth, and truth from epistemic status [4]. Fusion/Tunnel similarly treats AI outputs as knowledge-like candidates until checking and declared warrant criteria are satisfied [7]:

$$AI(Q) = K_{\text{like}}, \qquad K_{\text{like}} \neq K_{\text{validated}|V^*}.$$

A retained CTSA item can therefore be false, unsafe, misapplied, or poorly warranted. CTSA classifies the content of retained change; it does not certify knowledge.

5.3 Gain and loss are co-equal readouts

The programme's hypothesis-generation work introduced explicit defect accounting and a no-silent-loss discipline [5]. External evidence reinforces the need. Vives et al. show that representational differentiation can improve discrimination while impairing generalization [14]; Holton et al. show that reuse can support transfer while increasing interference [15]. A session can therefore add one capability while narrowing another.

Accordingly, CTSA does not define Human Return as "more items." It records a **Gain Profile** and a co-equal **Loss Profile**. Net interpretation is deferred until the loss, regulation, warrant, and direction audits are complete.

6. CTSA: Four Return Readout Classes

6.1 C - Conceptual Return

Definition. Conceptual Return occurs when the unaided human can reconstruct, discriminate, and appropriately use a concept or distinction that was not previously available in that usable form.

Human Learning already treats a word as a site where experience, memory, value, evidence, goals, constraints, and action converge, and builds meaning neighborhoods rather than treating vocabulary as inert storage [2]. Fusion/Tunnel treats articulation as a transport that can make distinctions nameable, comparable, and revisable [6]. External concept-learning work makes the compositional point stronger: existing concepts can become reusable building blocks for later concepts [12].

Conceptual Return is therefore not "one new word." It can be demonstrated when a person can independently:

- separate a distinction that was previously collapsed;
- state boundary conditions and misapplication cases;
- locate the concept in a relevant problem-context;

Table 2: External literature map and the residual role left for CTSA.

Literature	What is already established or strongly motivated	Constraint on CTSA
Educational taxonomy / SRL	Conceptual and procedural knowledge are established; revised Bloom includes metacognitive knowledge; SRL distinguishes declarative, procedural, and conditional strategy knowledge and emphasizes monitoring/control.	C/T/S are not new kinds of knowledge. Tool Return must be framed as context-appropriate selection; metacognition must be cross-cutting rather than silently omitted.
Concept learning / curriculum	Existing concepts can be reused and recombined to construct more complex concepts; divide-and-conquer curricula can facilitate later multi-cue performance.	Supports compositional learning and role-structured access, but does not validate CTSA-specific causal ordering.
GenAI education	AI can improve in-session performance while later unaided learning may fail; meta-analytic averages are heterogeneous and assessment quality matters.	Justifies the session-boundary return horizon and demands delayed/transfer measures.
Human-AI reliance	Confidence calibration, uncertainty expression, verification cost, and cognitive forcing alter reliance and checking behavior.	Warrant cannot be inferred from fluency; CTSA needs a metacognitive/regulation layer and exercisable resistance.
Human-AI complementarity	Average human augmentation does not imply synergy over the better component.	CTSA Human Return must remain separate from system-level augmentation and synergy.
AI and diversity	GenAI can improve individual creative output while reducing collective diversity.	Alternative Return cannot be raw idea volume; diversity and non-equivalence require independent coding.
Semantic / continual learning trade-offs	Greater representational separation can improve discrimination while harming generalization; reuse can facilitate transfer while increasing interference.	Gross gain is not net expansion; a co-equal Loss Profile is required.

- contrast the pre-AI and post-AI model without reconstructive help from the AI.

Conceptual Return ≠ vocabulary exposure. A new term that cannot change discrimination, representation, or problem-appropriate use after AI removal is lexical exposure, not demonstrated Conceptual Return.

6.2 T - Tool-Selection Return

Definition. Tool-Selection Return occurs when the unaided human can identify, select, and justify an external instrument, source, representation, collaborator, or access route appropriate to the declared problem-context.

This class deliberately does not claim a new knowledge type. Human Learning already treats tool selection as conditional knowledge: the same AI may be a generator, critic, calculator, or distraction depending on the task [2]. SRL literature likewise distinguishes knowing that a strategy exists from knowing how, when, and why it is applicable [9]. Tool-Selection Return is the Human-AI session-boundary operationalization of that conditional competence.

A strong Tool-Selection Return requires the human to answer:

- What kind of access does this problem require?
- Why is this tool appropriate here rather than merely available?
- What can the tool return, and what can it not establish?
- What independent route remains if the tool fails or conflicts with another source?

Readout Genesis keeps tools neutral: repositories, calculators, AI systems, and workflow software are not epistemic primitives [4]. Tool selection therefore opens a route to checking; it does not itself provide warrant.

6.3 S - Skill-Execution Return

Definition. Skill-Execution Return occurs when the human can execute a task-relevant procedure accurately enough to matter, without the AI performing the decisive operation, and can transfer that capability to a sufficiently new case.

Human Learning draws the relevant distinction already: workflow executed once is not skill; skill requires proceduralization and feedback-rich practice [2, 10, 11]. Human LoRA adds persistence and transfer as conditions for durable adaptation [3]. Bastani et al. provide a direct Human-AI warning: assisted success can coexist with worse unaided performance after AI removal [16].

Candidate Skill-Execution Returns include the ability to independently identify a confounder, trace provenance, construct a falsifier, compare study designs, detect an error, or apply the same reasoning pattern to a novel case.

Skill Return ≠ procedural imitation. Repeating an AI-demonstrated procedure on the same template is insufficient. The decisive evidence is unaided execution plus transfer under a declared criterion.

6.4 A - Alternative-Generation Return

Definition. Alternative-Generation Return occurs when the unaided human can access or generate a non-equivalent rival explanation, hypothesis, framing, or solution route that was not previously accessible in the declared task space.

This class is directly inherited from the programme's Difference requirement and Problem-to-Hypothesis architecture. Reachable and accessible are not identical; a candidate can exist structurally while remaining neglected because prior structure dominates search [5]. Fusion/Tunnel requires at least one non-equivalent distinction, rival, or route beyond paraphrase for expansion beyond the retained frame [6].

External evidence cautions against counting volume. Doshi and Hauser found that access to GenAI ideas improved judged individual creativity while outputs became more similar to one another [24]. Accordingly, twenty paraphrases of one causal story may yield no Alternative Return; one genuinely different causal route may.

Alternative Return $\neq$ idea count. Alternatives should be coded under declared equivalence rules by differences in consequence, explanation, intervention, prediction, or decision route.

7. Workflow and Metacognition: Cross-Cutting, Not Fifth Readouts

7.1 Where workflow belongs

Human Learning defines workflow as a repeatable organization of operations: steps, checkpoints, retrieval points, constraint tests, and tool-use decisions that make understanding inspectable and reusable [2]. Workflow is therefore a *process architecture*, not a return-content class.

A useful but non-mandatory relation is

$$C \to T \to \text{Workflow} \to S \to A \to C'.$$

The arrows are not a universal causal law. A new alternative may revise the concept; a new tool may reveal an alternative; a new skill may alter workflow. The purpose of the diagram is role separation.

7.2 Metacognitive regulation is a control axis

A global literature review exposes a second omission that should not be repaired by adding a fifth letter. Revised Bloom explicitly includes metacognitive knowledge [8]; SRL models treat planning, monitoring, strategy selection, evaluation, and control as central [9]. Human-AI studies add a specific reason: appropriate reliance depends partly on calibrated human self-confidence [19], and AI uncertainty expression can alter reliance and accuracy [20].

Metacognition is therefore represented as a cross-cutting regulation axis $\mathcal{M}$, asking whether the human can monitor uncertainty, choose when to use a CTSA gain, notice failure, and revise strategy. It governs the use of all four classes without being another content type.

CTSA asks what returned; metacognitive regulation asks whether the human can govern what returned.

8. Two Axes of Human Return

The Human Return Test and CTSA answer different questions and should be crossed, not substituted.

Axis I - What returned? CTSA classifies retained content/access as Conceptual, Tool-Selection, Skill-Execution, or Alternative-Generation Return.

Axis II - Can the human demonstrate that it returned? The existing Human Return battery tests reconstruction, discrimination, transfer, and independently generated next-question quality [6, 7].

Table 3: CTSA classes crossed with unaided demonstration.

Readout class	Unaided evidence
Conceptual	Reconstruct and correctly discriminate the new distinction; apply it under relevant boundary conditions.
Tool-Selection	Select an appropriate access route and explain why it fits the problem and what it cannot establish.
Skill-Execution	Execute the procedure and transfer it to a novel case without decisive AI assistance.
Alternative-Generation	Generate/discriminate a non-equivalent rival and state what observation or test would separate it.

A person may show strong Conceptual Return but weak Skill Return; Tool-Selection Return with no rival generation; Alternative Return without warrant; or Skill Return with poor metacognitive calibration. These profiles should remain visible rather than compressed into one "learning gain" score.

9. The Full Human-Return Audit

The final interpretation requires more than CTSA. A compact record is

$$\mathfrak{R}_H = \langle \mathcal{G}_{\text{CTSA}}, \mathcal{L}, \mathcal{M}, \mathcal{P}, \mathcal{W}, \Delta_{\text{dir}} \rangle.$$

The tuple is a bookkeeping architecture, not a calibrated universal score.

9.1 Gain Profile

What new CTSA items can the human demonstrate unaided relative to a frozen pre-AI baseline?

9.2 Loss Profile

What previously accessible distinctions, alternatives, uncertainty sensitivities, checking habits, or skills are now less available? The loss profile is co-equal with gain because discrimination, transfer, and interference can trade off [14, 15].

9.3 Metacognitive regulation

Can the person identify uncertainty, monitor error, decide when a strategy/tool applies, and change course? Confidence in self must remain distinct from trust in AI [19, 7].

9.4 Provenance

Where did the retained item come from: human-originated, AI-generated, external-record-derived, checker-derived, or mixed? Provenance supports auditability; it is not a truth certificate.

9.5 Warrant and exercisable resistance

What independent checking has the item survived? Vasconcelos et al. show that verification is behaviorally sensitive to task and explanation costs [21]; Buçinca et al. show that cognitive forcing can reduce overreliance by requiring more active engagement [22]. Therefore an available resistance route that a person will not or cannot exercise should not be treated as equivalent to effective checking.

9.6 Epistemic direction

Fusion/Tunnel already separates retention magnitude from retained direction: a persistent change can enlarge corrigibility or stabilize confirmation, same-basin attraction, outsourcing, or unwarranted certainty [7]. CTSA therefore cannot define "more" as "better."

10. Session-Boundary Architecture

The full programme can now be represented as a continuous sequence:

$$\begin{aligned}&\text{Frozen Human baseline} \rightarrow K_{\text{like}}\\ &\rightarrow \text{Difference / Resistance} \rightarrow H \leftrightarrow AI\\ &\rightarrow \text{Retention gate} \rightarrow \boxed{\text{CTSA Return Profile}}\\ &\rightarrow \text{Unaided Return Test} \rightarrow \text{Gain/Loss}\\ &\rightarrow \text{Regulation} \rightarrow \text{Provenance/Warrant}\\ &\rightarrow \text{Epistemic direction.}\end{aligned}$$

This location is conceptually important. CTSA is neither the beginning of inquiry nor the final truth test. It sits at the boundary where conversational residue is claimed to have become human capability. The architecture preserves the non-collapse laws developed across the programme:

- AI fluency $\neq$ human baseline;
- explanation $\neq$ verification;
- output count $\neq$ epistemic diversity;
- exposure $\neq$ retention $\neq$ improvement;
- trust in AI $\neq$ calibrated confidence in self;
- augmentation $\neq$ synergy $\neq$ Human Return;
- retained gain $\neq$ net expansion;
- provenance $\neq$ warrant;
- persistence $\neq$ truth.

Vaccaro et al.'s meta-analysis reinforces the system-level separation between augmentation and synergy [23]; Bastani et al. independently block the collapse between assisted performance and later human learning [16].

11. Worked Example: Checking an AI-Generated Causal Claim

Suppose a user asks whether a four-day work week "raises productivity by 40%." Before AI use, the person has a vague concept of evidence, no explicit distinction between correlation and causal effect, knows only generic web search as a checking tool, has no repeatable study-comparison skill, and can imagine only one explanation: fewer working days improve focus.

After a structured Human-AI inquiry and an unaided delay, a candidate return profile might be:

Conceptual Return: the user independently distinguishes correlation, causal effect, selection bias, measurement choice, and short-run versus long-run productivity.

Tool-Selection Return: the user chooses a primary report, longitudinal dataset, preregistered experiment, or domain-specific database according to the claim rather than treating a search summary as decisive evidence.

Skill-Execution Return: the user independently states the target causal claim, identifies plausible confounders, compares designs, and specifies evidence that would weaken the preferred explanation.

Alternative-Generation Return: the user generates non-equivalent rivals such as selection effects, Hawthorne effects, attrition, changed measurement, or baseline-performance differences.

The Human Return battery then tests whether those changes survive on a different causal claim. The Loss Profile asks whether the user became less tolerant of uncertainty or over-learned a single skeptical template. Metacognitive regulation asks whether the user knows when the template applies. Provenance separates AI-suggested rivals from independently recovered evidence. Warrant records which rivals remain merely possible and which have evidential support.

The example shows why CTSA is not a checklist of "things AI taught." It is a session-boundary readout of what became usable in the human and what remains open to correction.

12. Hypotheses and Legitimate Defeaters

The architecture generates open hypotheses; it does not protect them from removal.

H1 - Incremental Readout Hypothesis [Open]. A CTSA Return Profile will explain reproducible variance in delayed unaided human capability beyond ordinary post-test performance, recall, confidence, and established concept/skill measures.

H2 - Partial Dissociation Hypothesis [Open]. Conceptual, Tool-Selection, Skill-Execution, and Alternative-Generation returns will show partially dissociable profiles under matched interaction conditions rather than behaving as one interchangeable gain.

H3 - Return-vs-Exposure Prediction [Externally constrained]. Items visible only while AI support is present will predict assisted performance more strongly than delayed unaided capability. Bastani et al. provide direct evidence for this general separation, although not for CTSA classes specifically [16].

H4 - Gain/Loss Asymmetry Hypothesis [Open]. Positive CTSA gain can coexist with negative loss in previously accessible routes or checking habits; gross gain will therefore overestimate net human expansion. Semantic and continual-learning tradeoffs make this plausible without validating the CTSA-specific claim [14, 15].

H5 - Recombination Hypothesis [Open]. CTSA gains will sometimes interact compositionally: a new Concept may improve Tool selection; Tool plus Skill may make an Alternative accessible; a new Alternative may trigger conceptual restructuring. Conceptual bootstrapping and curriculum-learning results motivate but do not establish this dependency [12, 13].

H6 - Regulation/Direction Hypothesis [Open]. Equal gross CTSA gains will yield different later corrigibility depending on metacognitive calibration, effective resistance, provenance visibility, and whether the interaction reinforced or escaped a preexisting semantic basin.

Removal conditions. CTSA should be narrowed, merged, or abandoned if (i) independent coders cannot classify the four return classes reliably; (ii) the classes add no stable information beyond established conceptual/procedural/conditional/metacognitive measures; (iii) delayed CTSA profiles fail to predict or explain anything beyond ordinary performance and transfer; (iv) gain/loss coding is too task-dependent to be reproducible under declared scopes; or (v) a simpler rival architecture captures the same phenomena with fewer assumptions.

These are legitimate defeaters, not rhetorical threats. The programme's own methodological rule remains: an architecture that cannot lose variables when they stop doing explanatory work becomes vocabulary rather than theory [7].

13. Empirical Programme

The first study should not attempt to validate the whole Readout Programme. It should test the smallest claim that can make CTSA earn its place.

13.1 Primary question

> *Does a CTSA Return Profile add reliable, reproducible information about delayed unaided human capability beyond ordinary performance scores and established knowledge/skill measures?*

13.2 Minimum design

A preregistered study should include:

- a frozen pre-AI human baseline;
- a Human-only control and an AI-assisted condition;
- where feasible, an AI-only baseline for augmentation/synergy separation;
- an immediate assisted measure, followed by AI removal;
- at least one immediate unaided measure and one delayed novel-transfer measure at a preregistered domain-appropriate interval;
- tasks that permit concept discrimination, tool selection, skill execution, and rival generation;
- independent coding of gain, loss, provenance, and warrant;
- a metacognitive measure such as confidence calibration, uncertainty monitoring, or strategy-choice accuracy.

13.3 Candidate outcomes

- **Conceptual:** correctly reconstructed distinctions; boundary-condition accuracy; misapplication rate.
- **Tool-Selection:** appropriate selection accuracy; justification quality; rejection of an inappropriate but salient tool.
- **Skill-Execution:** unaided procedural accuracy; error detection; self-correction; transfer to structurally related novel tasks.
- **Alternative-Generation:** number of readout-distinct rival classes; discriminability; ability to state what evidence separates them.
- **Loss:** decline in previously available alternatives, generalization, uncertainty sensitivity, checking behavior, or baseline skill.
- **Regulation:** confidence calibration, monitoring accuracy, and context-appropriate strategy/tool choice.

13.4 Analysis priority

The decisive analysis is incremental validity, not whether all four CTSA scores rise. Models should compare delayed human outcomes predicted by ordinary performance/recall measures alone versus the same models augmented by CTSA, Loss, and regulation variables. Cross-validation or held-out replication should be preferred to post-hoc fit. If CTSA adds no stable explanatory or predictive value, the taxonomy has not earned its complexity.

14. Adversarial Cases

14.1 Vocabulary illusion

A user learns ten technical terms and repeats their definitions. If no new distinction can be applied, no tool selected, no skill transferred, and no rival generated, this is lexical exposure rather than demonstrated human expansion.

14.2 Tool-name illusion

A user can name PubMed, a calculator, a causal diagram, and a database but cannot choose among them for a new problem. Recognition without conditional selection is not strong Tool-Selection Return.

14.3 Procedural imitation illusion

A user reproduces the exact workflow demonstrated with AI but fails on a structurally related case. The interaction may have produced local imitation rather than Skill-Execution Return.

14.4 Alternative-volume illusion

An AI produces twenty differently worded explanations from the same causal frame. Without non-equivalent consequences or discriminating tests, the apparent diversity collapses to one Alternative class.

14.5 Gain-with-loss tunnel

A user gains new concepts and a fast procedure but becomes less willing to check evidence, less sensitive to uncertainty, or less able to generate rivals independently. Gross CTSA rises while epistemic direction deteriorates. The loss and direction audits are therefore constitutive to interpretation.

14.6 Metacognitive blind gain

A person acquires a real skill but cannot recognize when it is inappropriate, becomes overconfident, and applies it indiscriminately. Skill Return can be positive while metacognitive regulation is negative. CTSA alone must not call this net expansion.

15. Claim Boundaries

This paper does **not** claim that:

- CTSA is an exhaustive taxonomy of cognition, learning, intelligence, expertise, or wisdom;
- C, T, S, and A are biological modules, literal neural variables, or four new kinds of knowledge;
- all four classes are necessary for every learning goal;
- more CTSA items always imply more human potential;
- Tool-Selection Return is conceptually distinct from all established conditional or strategic knowledge constructs;
- Skill-Execution Return replaces established procedural-learning or expertise theories;
- retained alternatives are true merely because they are diverse;
- a retained concept, tool, skill, or alternative is beneficial merely because it persists;
- metacognitive regulation can be reduced to self-confidence alone;
- AI is the sole cause of post-session human change;
- the proposed readouts already possess validated universal scales;
- Human Return should define success where durable human capability is outside the declared goal.

The strongest permitted claim is architectural: CTSA proposes a session-boundary method for keeping several questions separate that Human-AI research can otherwise collapse—what changed in the human, whether it persisted, what was lost, whether the person can regulate it, where it came from, what checking it survived, and whether the retained direction expands or narrows corrigible inquiry.

16. Standalone Thesis and Programme Continuity

> **CTSA does not propose four new kinds of human knowledge. It proposes a session-boundary measurement architecture for distinguishing what kind of capability, if any, has actually returned to the human after AI assistance is removed. Conceptual Return identifies a new usable distinction; Tool-Selection Return identifies a newly usable access route; Skill-Execution Return identifies unaided transferable performance; and Alternative-Generation Return identifies a non-equivalent route the human can independently reach. These gains count toward human expansion only when persistence, losses, metacognitive regulation, provenance, warrant, and epistemic direction remain separately auditable.**

The thesis continues the internal programme without replacing it. *When AI Expands Human Potential* supplies the normative target; *Human Learning as Epistemic Architecture* supplies the movement from meaning to corrigible competence; *Experience Is the Human LoRA* supplies selective persistence; Readout Genesis supplies retained-difference, translation, and loss discipline; *From Problem to Hypothesis* supplies bounded semantic mobility and discriminable alternatives; and *Epistemic Fusion or Tunnel* supplies reciprocal dialogue, candidate status, resistance, retention, direction, and the unaided Human Return horizon. CTSA is the measurement interface placed exactly where those strands meet: the moment a conversational residue is claimed to have become a human capability.

The next step is deliberately modest. CTSA should not be defended by adding more terminology. It should be tested against simpler measures, established learning constructs, and rival models. If it adds stable information about delayed unaided capability, it earns a role. If it does not, the programme should remove or compress it.

References

Programme lineage

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